Certified, Risk-Tiered Object Scheduling for Roadside Cooperative Perception over Measured C-V2X Sidelink Service

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***Abstract*—A roadside unit (RSU) that shares its perception with connected vehicles must decide, every 100 ms, which perceived objects to put in its collective perception message (CPM), at which level of detail, over a sidelink whose service varies and can only be partly predicted, while its own safety traffic must still be served. Standardized CPM generation rules decide by object dynamics, value-of-information schemes rank objects by generic relevance, and learned cooperative-perception models treat bandwidth as given; none of them guarantees that the objects a vehicle cannot see itself will arrive before their deadline. We propose CERTISCHED, a scheduler that (i) values an update by the error it removes from the receiving vehicle's world model, weighted by whether the vehicle can see the object and whether it is on a conflicting path, both estimated from information the RSU actually has; (ii) admits the most critical updates against an adaptive-conformal lower bound on the bytes the measured sidelink will deliver before the deadline, which yields a distribution-free long-run guarantee on the deadline-miss frequency even under service drift; (iii) prices any further critical traffic with a drift-plus-penalty virtual queue that enforces a time-average delivery contract for the RSU's background services; and (iv) fills the remaining service with best-effort updates that strict priority prevents from harming either class. We evaluate on real perception sequences from V2X-Seq-SPD and real C-V2X service measured in SEE-V2X, with a physical priority-queue replay, ground-truth criticality that the scheduler never sees, standard CPM sizes and contract-compliant baselines. On [run] measured links and [run] road sequences, CERTISCHED raises the awareness of hidden, conflicting objects to [run] versus [run] for ETSI CPM rules and [run] for a max-weight age-of-information policy, keeps the certified deadline-miss frequency at [run] for a target of 0.05, and meets the background contract ([run]) at [run] ms per decision.**

***Index Terms*—Collective perception, C-V2X sidelink, conformal prediction, age of information, deadline guarantees, Lyapunov optimization, roadside units.**

I. INTRODUCTION

AN infrastructure sensor at an intersection sees what a connected vehicle (CV) cannot: a cyclist behind a parked truck, a car approaching from an occluded arm. Sharing that view through the collective perception service [1], [2] turns roadside sensing into a safety benefit only if the right objects reach the vehicle before they become stale. The RSU therefore faces a recurring scheduling decision. Every 100 ms it holds tens of tracked objects, each of which it can describe briefly (position and velocity) or fully (box, heading, class), possibly with a blind repetition; the sidelink will deliver an unknown number of bytes within the deadline; and the same radio also carries the RSU's own safety services, such as basic safety and signal phase messages, which must keep their delivery rate.

Existing approaches each address part of this decision. ETSI CPM generation rules include an object when it has moved, changed speed or heading, or when a timer expires [3], [4]; redundancy-mitigation and value-of-information (VoI) schemes rank objects by novelty, accuracy or distance [5]–[9], and the current revision of the standard moves toward VoI-ordered inclusion under a resource-dependent object count [8]. Age-of-information (AoI) scheduling keeps every source fresh [10]–[12]. Learned cooperative perception compresses features under an exogenous bandwidth [13]–[17]. Three gaps remain. First, relevance is rarely receiver-centric: an object the vehicle already sees is worth little, an object it cannot see on a colliding path is worth a lot, and the RSU can estimate both. Second, the delivered service of a real sidelink drifts with load, distance and interference [18], [19]; a budget set from a point forecast is exceeded in roughly half of the intervals, while a fixed conservative budget wastes capacity, and no existing scheduler states what deadline reliability it actually provides. Third, safety traffic of the RSU itself must not be starved by perception traffic, which calls for a long-run contract rather than a fixed share.

This paper closes these gaps with CERTISCHED, an RSU-side scheduler with a certified deadline tier. Our contributions are:

1) *Receiver-centric problem formulation.* We model the value of an object update as the expected reduction of the receiving vehicle's awareness loss: the error of its extrapolated estimate (with state expiry, unacknowledged broadcast and acceleration-driven uncertainty growth), weighted by the vehicle's occlusion and path conflict and by the fidelity of the transmitted representation. Selection is a multiple-choice knapsack with a risk-tiered chance constraint and a time-average background constraint (Section III).
2) *Certified scheduling.* We bound the bytes the sidelink will deliver before the deadline with adaptive conformal inference [20] on lagged sidelink measurements and prove that, under strict priority, the long-run frequency of deadline misses of the certified tier is at most the target α plus $O(1/T)$ for *any* service sequence (Theorem 1). A drift-plus-penalty queue [21] admits further critical traffic while meeting the background contract (Theorem 2); best-effort updates use the residual service without affecting either guarantee (Section IV).

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3) *A measurement-grounded evaluation protocol.* We combine real roadside perception sequences [22] with real C-V2X service traces [18] through a physical priority-queue replay, use ground-truth criticality that no policy observes, standard CPM sizes with representation modes that change what the receiver reconstructs, and compare against ETSI CPM rules, VoI redundancy mitigation, max-weight AoI, confidence-density selection and full transmission under the same service contract (Section V).
4) *Results.* Across [run] paired episodes, CERTISCHED delivers higher awareness of hidden, conflicting objects than every contract-compliant baseline while its certified miss frequency tracks the target and the background contract holds; adaptive conformal envelopes keep calibration under drift where split and fixed-level conformal envelopes do not (Section VI).

II. RELATED WORK

A. Communication in Cooperative Perception

Real-world benchmarks such as DAIR-V2X [23], V2X-Seq [22] and RCooper [24] enabled cooperative perception with roadside sensors [25]. Learned systems decide what to share inside the network: Who2com and When2com learn whom to request from [16], [17]; V2VNet and V2X-ViT fuse intermediate features [26], [27]; Where2comm, How2comm, CodeFilling and TransIFF select spatial regions, instances or codes under a bandwidth budget [13]–[15], [28]. Recent work allocates radio resources for collaborative perception with learning [29], [30] or joint optimization in simulation [31], [32]. These methods take the budget as given and optimize detection accuracy; they do not provide deadline reliability under measured service, and most assume feature-level sharing, whereas deployed RSUs broadcast object-level CPMs.

B. CPM Generation, Redundancy and Value of Information

The ETSI collective perception service includes an object when its position changes by more than 4 m, its speed by more than 0.5 m/s, its heading by more than 4°, or when 1 s (0.5 s for vulnerable road users) has passed [1], [3]. Thandavarayan *et al.* analyzed these rules and proposed redundancy mitigation and look-ahead inclusion [4], [5]; Delooz and Festag adapted CPM load to the channel [6]. VoI formulations weight objects by accuracy, relevance and novelty [7], [9], [33], and the standard now favors VoI-ordered inclusion with a resource-dependent number of objects [8]. Our value function belongs to this family but is defined from the receiver's world model (Section IV-A), and our admission adds a reliability guarantee that none of these schemes provides.

C. Freshness: AoI and Age of Incorrect Information

AoI measures the time since the latest received update was generated [10], [12]; max-weight and Whittle-index policies minimize weighted AoI over unreliable broadcast channels [11]. The age of incorrect information penalizes only time spent with a wrong estimate [34], and AoI–cost trade-offs have been studied for perception messages on the NR-V2X sidelink [35]. Our estimation-error term is an AoII-type measure with an explicit uncertainty growth, so it degenerates to AoI-like refresh for objects whose motion is unpredictable.

D. Sidelink Service and Its Uncertainty

In LTE-V2X Mode 4 every station selects resources autonomously by sensing-based semi-persistent scheduling [36], [37]; packet delivery depends on distance, half-duplex and collisions [19], and congestion control limits each station by the channel busy ratio [38], [39]. SEE-V2X released application traces and sidelink control information (SCI) captured with commercial radios [18], which exposes the delivered service an RSU would actually face. Probabilistic service curves from stochastic network calculus bound delays under fading [40], [41] but require a channel model. Conformal prediction gives distribution-free coverage [42], [43], has been applied to wireless calibration [44], and adaptive variants keep long-run coverage under arbitrary distribution shift [20], [45]. We use this property to certify a deadline tier directly from measurements.

E. Long-Run Constraints and Slicing

Lyapunov drift-plus-penalty methods enforce time-average constraints through virtual queues [21], [46]. Safe reinforcement learning for RAN slicing projects actions onto latency-feasible sets [47]; it targets base-station slicing rather than sidelink broadcast of object messages.

Table I positions CERTISCHED: it is, to our knowledge, the first object-level V2X scheduler with a distribution-free deadline guarantee calibrated online from sidelink measurements.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Entities and Timing

An RSU with a roadside LiDAR serves a CV through the sidelink (Fig. 1). Time is divided into decision epochs $t = 1, 2, \ldots$ of length $\Delta = 100$ ms aligned with the sensor frames. At epoch t the RSU holds tracked objects $\mathcal{I}_t$ with world-frame state $s_{i,t} = (x, y, v_x, v_y, l, w, h, \psi, \text{class})$ captured at $c_{i,t}$, and the latest CV pose p_t and velocity u_t received in a cooperative awareness message. After a processing delay δ, the RSU enqueues CPM segments. An update is useful if it arrives within D of capture, i.e., within the relative deadline $\tau = D - \delta$ of the decision.

B. Message Representations

Each object can be sent in a mode $m \in \mathcal{M}$ (Table II). Sizes follow the CPM structure [1] with the per-message and per-object sizes used in the CPM literature [4]. The *kinematic* mode omits the box, so the receiver reconstructs it from a class prior and the heading of motion; the *repetition* mode adds a blind retransmission. Modes therefore change what the receiver knows, and the evaluation recomputes IoU from the transmitted content. A segment carries at most K objects and one header of H bytes.

TABLE I
POSITIONING AGAINST REPRESENTATIVE APPROACHES

Approach	Unit	Receiver-centric value	Deadline guarantee	Service model	Background protection	Evidence
ETSI CPM rules [1], [4]	object	no (dynamics)	no	none	DCC	simulation
VoI / redundancy mitigation [5], [7]	object	partial (distance)	no	resource count	no	simulation
Max-weight AoI [11]	source	no	no	i.i.d. erasure	no	analysis
Where2comm / CodeFilling [13], [15]	feature	no	no	fixed budget	no	datasets
ITORA / FullPerception [31], [32]	feature/region	partial	no	simulated	resource constraint	simulation
CERTISCHED (ours)	object + mode	occlusion, conflict, error	long-run, distribution-free	measured, adaptive conformal	time-average contract	real traces

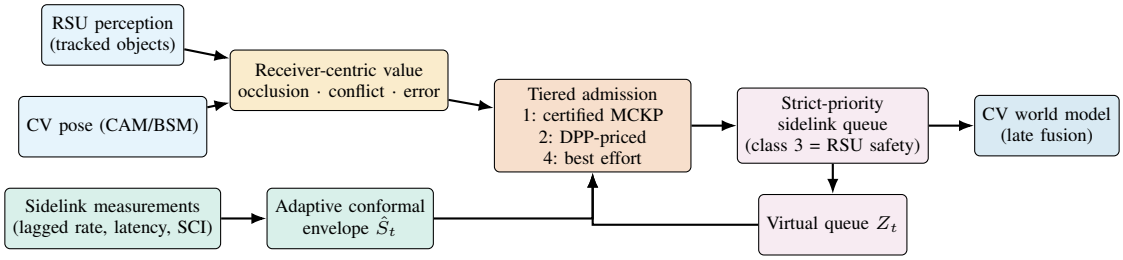


Fig. 1. CERTISCHED at the RSU. Decisions use only RSU tracks, the CV's broadcast pose and lagged sidelink measurements; ground truth is used only offline for evaluation.

TABLE II
CPM REPRESENTATION MODES

Mode	Bytes/object	Receiver reconstructs	Copies
kinematic	35	position, velocity; class-prior box	1
box	60	position, velocity, box, heading	1
box + repetition	2×60	as box	2
Header $H = 121$ B per segment, at most $K = 10$ objects per segment			

C. Sidelink Service Model

In window t the sidelink serves the RSU at a gross rate r_t (bytes/ms); each packet experiences an access latency $\ell \le L_t$ drawn from the measured latency distribution of the window and is erased independently with probability p_t. Packets are served by strict priority in four classes: (1) certified critical updates, (2) priced critical updates, (3) the RSU's background safety traffic of A_t bytes, and (4) best-effort updates. Bytes not served by the end of the window are superseded by the next CPM. The bytes deliverable before the deadline at the head of the queue are

$$S_t(\tau) = r_t \, (\tau - L_t)^+. \tag{1}$$

Lemma 1 (Priority delivery)**.** *If the class-1 segments of epoch t total $B_t^{(1)} \le S_t(\tau)$ bytes (headers included) and $\tau \le \Delta$, every class-1 segment is served and arrives within τ of the decision.*

Proof: Class 1 is served first from the start of the window at rate r_t, so its last byte is served at $B_t^{(1)}/r_t \le \tau - L_t \le \Delta$, i.e., inside the window; adding an access latency $\ell \le L_t$ gives arrival at most τ. Packets of earlier epochs do not precede it because they are superseded at the window end. ∎

D. Receiver Model and Awareness

The CV fuses its own detections with received objects (late fusion). For each track it keeps the latest received state, extrapolates it at constant velocity and discards it after $T_{\text{exp}} = 1$ s. A ground-truth object is *aware* at time t' if an unexpired received state lies within $d_{\text{m}} = 2$ m of it. Objects that the vehicle cannot see itself are *hidden*; hidden objects whose future path comes within 5 m of the CV's future path within 5 s are *critical*. These labels come from ground truth and are used only for evaluation.

E. Problem Formulation

Let $x_{i,m,t} \in \{0,1\}$ select object i in mode m, with value $v_{i,m,t}$ (Section IV-A) and bytes b_m. Let $\mathcal{C}_t \subseteq \mathcal{I}_t$ be the objects the RSU estimates to be critical, S_t^{bg} the background bytes served, and $M_t \in \{0,1\}$ indicate that a certified update of epoch t misses its deadline for lack of service. The RSU seeks

$$\max_{x} \ \liminf_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \sum_{i,m} v_{i,m,t} \, x_{i,m,t} \tag{2}$$

$$\text{s.t.} \ \sum_m x_{i,m,t} \le 1, \tag{3}$$

$$\limsup_{T\to\infty} \tfrac{1}{T} \textstyle\sum_t M_t \le \alpha, \tag{4}$$

$$\liminf_{T\to\infty} \tfrac{1}{T} \textstyle\sum_t \left(S_t^{\text{bg}} - \rho A_t\right) \ge 0, \tag{5}$$

where every decision may use only information available at epoch t, and $S_t(\tau)$ is revealed only after the window. Constraint (4) is a chance constraint stated over time, which is the natural reliability notion for a non-stationary channel; (5) is the background delivery contract with ratio ρ.

IV. CERTISCHED

A. Receiver-Centric Value of an Update

Ego relevance: The RSU casts five sight lines from p_t to the centre and shrunken corners of each object and counts those blocked by a tall (≥ 1.2 m) object nearer to the CV; the blocked fraction is the occlusion estimate $o_{i,t} \in [0,1]$, set to 1 beyond the CV sensor range R. With both trajectories extrapolated at constant velocity, the closest approach over

horizon H_c has distance $d_{i,t}^{\min}$, and $\kappa_{i,t} = e^{-d_{i,t}^{\min}/\sigma_d}$ measures conflict. The relevance weight is

$$w_{i,t} = w_0 + (1 - w_0)\, o_{i,t}\big(k + (1-k)\kappa_{i,t}\big), \tag{6}$$

so a visible object keeps a small weight w_0, a hidden object at least $w_0 + (1-w_0)k$, and a hidden conflicting object close to 1. Object i is estimated critical ($i \in \mathcal{C}_t$) if $o_{i,t} \geq o^\star$ and $d_{i,t}^{\min} \leq d^\star$.

Receiver error: Let $\hat{s}_i$ be the last state the RSU transmitted and π_i the probability that it was received (broadcast has no acknowledgement). At the deadline $t+\tau$ the receiver's extrapolation differs from the RSU's prediction by $e_{i,t}^{\mathrm{k}}$; unmodelled acceleration $a_{\max}$ adds $\frac{1}{2}a_{\max}\theta^2$ over the belief age θ. We use

$$e_{i,t} = \sqrt{(e_{i,t}^{\mathrm{k}})^2 + (\tfrac{1}{2}a_{\max}\theta^2)^2}, \qquad e_{i,t} = \infty \text{ if } \theta \geq T_{\exp}, \tag{7}$$

and the expected awareness loss $\bar{h}_{i,t} = \pi_i\big(1 - e^{-e_{i,t}/\epsilon}\big) + (1 - \pi_i)$.

Representation fidelity: For the box modes $q_{i,m} = 1$; for the kinematic mode $q_{i,m}$ is the bird's-eye-view IoU between the RSU box and its class-prior reconstruction, computed per object.

The value of sending i in mode m under the lagged loss estimate $\hat{p}_t$ is

$$v_{i,m,t} = w_{i,t}\,\bar{h}_{i,t}\,\big(1 - \hat{p}_t^{c_m}\big)\big(\eta + (1-\eta)q_{i,m}\big). \tag{8}$$

All terms use RSU-side information only.

B. Adaptive Conformal Service Envelope

The scheduler needs a lower bound $\hat{S}_t$ on $S_t(\tau)$ before window t. A quantile gradient-boosting model f maps lagged features ϕ_t (rate, latency, loss, channel busy ratio and SCI statistics of windows $t-1, t-2, t-3$ and 10-window summaries) to $S_t(\tau)$; it is trained on other scenarios only (leave-one-scenario-out). With nonconformity $e_u = f(\phi_u) - S_u(\tau)$ and the empirical $(1-\alpha_t)$-quantile $Q_{1-\alpha_t}$ of past scores,

$$\hat{S}_t = \begin{cases} 0, & \alpha_t \leq 0 \\ \big(f(\phi_t) - Q_{1-\alpha_t}(\{e_u\}_{u<t})\big)^+, & 0 < \alpha_t < 1 \\ +\infty, & \alpha_t \geq 1 \end{cases} \tag{9}$$

and after observing the window, $\alpha_{t+1} = \alpha_t + \gamma\big(\alpha - \mathrm{err}_t\big)$ with $\mathrm{err}_t = \mathbb{1}\{S_t(\tau) < \hat{S}_t\}$ [20]. The level rises after covered windows and falls after violations, which re-calibrates the envelope when the channel drifts.

C. Tiered Admission

Algorithm 1 runs each epoch. Class 1 solves the multiple-choice knapsack (MCKP)

$$\max \sum_{i \in \mathcal{C}_t, m} v_{i,m,t} x_{i,m,t} \;\text{ s.t. }\; \sum_{i,m} b_m x_{i,m,t} + n_t H \leq \hat{S}_t, \tag{10}$$

where $n_t \in \{1, 2\}$ headers account for a repetition segment. Critical objects that do not fit are offered to class 2, whose MCKP maximizes $V\sum v\,x - Z_t \sum b\,x$ under the point forecast; the virtual queue

$$Z_{t+1} = \big[Z_t + \rho A_t - S_t^{\mathrm{bg}}\big]^+ \tag{11}$$

Algorithm 1 CERTISCHED decision at epoch t

Require: tracks $\mathcal{I}_t$, CV pose, lagged measurements, Z_t, α_t

1: compute $o_{i,t}, \kappa_{i,t}, e_{i,t}, q_{i,m}$ and $v_{i,m,t}$ by (8); split $\mathcal{C}_t$
2: $\hat{S}_t \leftarrow$ envelope (9); $\bar{S}_t \leftarrow$ point forecast
3: class 1: solve (10) on $\mathcal{C}_t$ (exact DP) $\rightarrow \mathcal{X}^{(1)}$
4: class 2: on $\mathcal{C}_t \setminus \mathcal{X}^{(1)}$ maximize $Vv - Z_t b$ within $\bar{S}_t \rightarrow \mathcal{X}^{(2)}$
5: class 4: plan remaining objects within $\bar{S}_t - A_t$; append the rest by v/b
6: enqueue CPM segments; observe $S_t(\tau)$ and S_t^{bg}
7: update α_{t+1} and Z_{t+1} by (11)

grows when the background is under-served. All remaining objects are planned in class 4 against the point forecast and then appended in value-per-byte order; strict priority lets them use only service left after classes 1–3.

D. Guarantees

Theorem 1 (Certified deadline tier)**.** *For any sequence of service values $\{S_t(\tau)\}$, any base predictor f, $\alpha, \alpha_1 \in (0,1)$ and $\gamma > 0$,*

$$\frac{1}{T}\sum_{t=1}^{T} M_t \leq \alpha + \frac{\max(\alpha_1, 1-\alpha_1) + \gamma}{\gamma T}.$$

Proof: Class-1 admission enforces $B_t^{(1)} \leq \hat{S}_t$. If $S_t(\tau) \geq \hat{S}_t$, Lemma 1 delivers every class-1 segment in time; hence $M_t \leq \mathrm{err}_t$. Under (9), $\alpha_t \leq 0$ gives $\hat{S}_t = 0 \leq S_t(\tau)$ and $\mathrm{err}_t = 0$, while $\alpha_t \geq 1$ gives $\mathrm{err}_t = 1$; therefore $\alpha_t \in [-\gamma, 1+\gamma]$ for all t by induction. Summing the update, $\alpha_{T+1} = \alpha_1 + \gamma\sum_{t\leq T}(\alpha - \mathrm{err}_t)$, so $\frac{1}{T}\sum_t M_t \leq \frac{1}{T}\sum_t \mathrm{err}_t = \alpha + \frac{\alpha_1 - \alpha_{T+1}}{\gamma T} \leq \alpha + \frac{\alpha_1 + \gamma}{\gamma T}$, which is at most the stated bound. ∎

Remark 1. The bound holds per sample path, without exchangeability or stationarity, which suits a sidelink whose load and geometry drift. Erasures are not service misses; with independent erasures a certified update in mode m is additionally lost with probability $p_t^{c_m}$, which the repetition mode reduces.

Assumption 1. *(r_t, L_t, A_t) are i.i.d. over epochs with bounded bytes, and with class-2 traffic switched off the background contract holds with slack $\epsilon > 0$: $\mathbb{E}[S_t^{\mathrm{bg}} - \rho A_t] \geq \epsilon$.*

Theorem 2 (Background contract)**.** *For every sample path, $\frac{1}{T}\sum_{t\leq T}(\rho A_t - S_t^{\mathrm{bg}}) \leq Z_{T+1}/T$. Under Assumption 1, class-2 admission by $\max V\sum vx - Z_t\sum bx$ keeps $\mathbb{E}[Z_t] = O(V/\epsilon)$, so* (5) *holds, and the class-2 value is within B/V of the best stationary class-2 policy, where B bounds the second moments of the per-epoch bytes.*

Proof: From (11), $Z_{t+1} \geq Z_t + \rho A_t - S_t^{\mathrm{bg}}$; summing gives the first claim since $Z_1 = 0$. Because background is served after classes 1–2 within the window, $S_t^{\mathrm{bg}} \geq S_t^{\mathrm{bg},0} - B_t^{(2)}$, where $S_t^{\mathrm{bg},0}$ is the background service without class 2. The one-step drift $\frac{1}{2}(Z_{t+1}^2 - Z_t^2)$ is therefore at most $B + Z_t(\rho A_t - S_t^{\mathrm{bg},0}) + Z_t B_t^{(2)}$, and subtracting $V\times$value gives the drift-plus-penalty expression whose per-epoch minimizer is the class-2

rule. The remaining steps are those of [21, Thm. 4.8] with the slack of Assumption 1. ∎

Proposition 1 (Solvers)**.** *Rounding byte costs up to a grid g, the dynamic program solves each MCKP exactly in $O(|\mathcal{I}_t||\mathcal{M}|\lceil \hat{S}_t/g \rceil)$ time without overstating feasibility. The LP-dominance greedy with the best-single-item safeguard attains at least half of the optimum in $O(|\mathcal{I}_t||\mathcal{M}|\log(|\mathcal{I}_t||\mathcal{M}|))$ [48], [49].*

V. Evaluation Methodology

A. Data

Perception: V2X-Seq-SPD [22] provides synchronized roadside and vehicle LiDAR sequences at real intersections with infrastructure-side, vehicle-side and cooperative 3-D labels, track identities and calibration. We use all [run] sequences ([run] infrastructure frames, [run] vehicle frames). The RSU's perception is the infrastructure-side track annotation, which isolates the scheduling question from detector quality; velocities are causal finite differences. The CV pose comes from its localization. Cooperative labels whose source is the infrastructure only are hidden; among them, [run] object-frames are critical by their future paths ([run] hidden object-frames in total).

Sidelink service: SEE-V2X [18] provides receiver traces and SCI from commercial LTE-V2X radios in perception-only and BSM-plus-perception applications, outdoors at an intersection and indoors over extended durations. Each receiver file is a link; per 100-ms window we extract delivered bytes, loss, channel busy ratio, latency quantiles and SCI statistics ([run] links in [run] scenarios, [run] windows). The gross service rate is the delivered bytes divided by the window's delivery ratio. Outdoor intersection links form the primary evaluation; indoor links are reported as a secondary condition.

Pairing: The datasets were not captured together. For each sequence we draw 8 outdoor and 4 indoor link segments with seeded hashes, starting after the envelope's calibration prefix; every policy in a block sees the same objects, CV poses, service, latency draws and erasure draws.

B. Replay

The replay implements Section III-C: CPM segments enter the priority queue, are served as a fluid at the window's rate, arrive after a latency drawn from the window's measured quantiles, and are erased with the measured loss. Unserved bytes cost no service and are dropped at the window end. The service of a window is exogenous: the counterfactual RSU may use at most the service that the measured flow obtained, which is conservative where the measured application did not saturate the channel; a CPM-size multiplier sweeps scarcity.

C. Policies

All baselines run in class 4, after the RSU's background traffic, so that they meet the same contract; their budget is the same point forecast that CertiSched uses for its best-effort tier. *ETSI CPM rules* include objects by the TS 103 324 triggers and order them by time since last inclusion [1]. *VoI redundancy mitigation* ranks by confidence × distance relevance × novelty [7]. *Max-weight AoI* ranks by $(1-\hat{p}_t)\theta(\theta+\Delta)$ [11]. *Confidence density* ranks by confidence per byte, the object-level analogue of Where2comm [13]. *Send all* transmits every object in box mode. As sensitivity, AoI and VoI also run in class 2, ahead of the background. *Oracle service* is CertiSched with $\hat{S}_t = S_t(\tau)$ (not causal). Ablations remove one component at a time (Table VI).

TABLE III
Parameters (all varied in sensitivity studies where noted)

Epoch Δ, processing δ, deadline D (τ)	100, 20, 100 (80) ms
Certified target α (sweep)	0.05 (0.01–0.20)
ACI step γ, score window, calibration prefix	0.005, 2000, 20%
Background contract ρ, DPP V (sweep)	0.95, 10^6 (10^4–10^7)
Background traffic A_t	2 streams × 300 B per 100 ms
CV sensor range R, conflict horizon H_c, σ_d	60 m, 5 s, 5 m
Critical estimate $o^\star$, $d^\star$	0.5, 8 m
w_0, k, ϵ, η, $a_{\max}$, $T_{\exp}$	0.05, 0.3, 1 m, 0.5, 1.5 m/s^2, 1 s
CPM sizes, segment size K (load sweep)	Table II, 10 (×1, 2, 4)

[run] figure data `envelope_calibration.csv`

Fig. 2. Controller-free envelope evaluation on all held-out windows of the measured links: (a) empirical miss frequency versus target; (b) mean bound relative to mean deliverable bytes.

D. Metrics and Statistics

Primary: critical awareness and hidden awareness, i.e., the fraction of (object, vehicle frame) pairs that are aware. Secondary: AoI of aware critical objects, AP of the fused object list at BEV IoU 0.5 and 3-D IoU 0.7 (car classes), background delivery ratio $\sum S^{\mathrm{bg}}/\sum A$, sidelink load, the certified miss frequency of Theorem 1 (epochs with a class-1 service miss over all epochs), and decision time. The road sequence is the statistical unit: run outcomes are averaged per sequence and 95% intervals use 5,000 cluster-bootstrap resamples [50]. The confirmatory family, fixed before the data run, compares CertiSched with each contract-compliant baseline on the two primary metrics (Wilcoxon signed-rank on sequence means [51], Holm correction [52]). Table III lists all parameters.

VI. Results

A. Calibration of the Service Envelope

Fig. 2 evaluates the envelopes on every held-out window, without any controller. The point forecast is exceeded in [run] of windows and the last-value rule in [run], so a budget set from a forecast fails about half of the time. For $\alpha = 0.05$ the adaptive envelope misses in [run] of windows, compared with [run] for split conformal and [run] for fixed-level rolling conformal, and it retains a fraction [run] of the mean deliverable bytes.

B. Main Comparison

Table IV reports the primary (outdoor) condition and Table V the confirmatory tests. CertiSched reaches a critical awareness of [run] and a hidden awareness of [run], against

[run] and [run] for ETSI CPM rules, [run] and [run] for VoI redundancy mitigation, and [run] and [run] for max-weight AoI. The mean age of the critical objects the vehicle knows is [run] ms (ETSI: [run] ms). The certified miss frequency is [run] for a target of 0.05, and the background delivery ratio is [run]. The oracle variant, which knows each window's service, reaches [run], which bounds what better service prediction alone could add.

C. Ablations

Table VI removes one component at a time. Replacing the conformal envelope by the point forecast raises the certified miss frequency to [run]; split and fixed-level conformal keep the average level but lose calibration where the link drifts (Fig. 2). Removing ego relevance or tiers lowers critical awareness to [run] and [run]; removing the best-effort tier lowers hidden awareness to [run]; removing the DPP queue changes the background ratio to [run].

D. Trade-offs and Scarcity

Fig. 3 shows the effect of the DPP parameter V, the certified target α and channel scarcity. Larger V admits more priced critical traffic at the expense of background delivery, as Theorem 2 predicts. The certified miss frequency stays below each target α (Theorem 1). As CPM sizes grow, every policy loses awareness; the gap between CERTISCHED and the baselines indicates how much of the scarce service each policy spends on objects the vehicle needs.

E. Secondary Condition and Computation

On indoor extended-duration links (Table VII) service is richer and steadier, so differences between policies shrink. A decision takes [run] ms on average in single-threaded Python on a laptop CPU with the exact dynamic program, far below the 100-ms epoch; the greedy variant (Proposition 1) is faster with a guaranteed half-optimal objective.

VII. Discussion and Limitations

What the guarantee covers: Theorem 1 bounds service-induced misses of the certified tier over time for any service sequence. It does not bound erasures, which the repetition mode reduces under an independence assumption, and it does not promise that every critical object is certified: when the envelope is small, the remaining critical objects go to class 2 or class 4 without a guarantee, and the tables report their outcomes.

Service-capped replay: The counterfactual RSU never uses more service than the measured application obtained in that window, which is conservative when the application did not saturate the channel. The datasets were not captured together, so road geometry does not influence the channel; the sweeps over scarcity and the two measurement conditions bound this effect but do not remove it. Over-the-air validation with commercial radios is the next step.

Perception: Using infrastructure annotations as RSU perception isolates scheduling from detection errors; a learned detector would add false and missed objects that affect all policies alike, and the value function accepts detector confidence through π_i and the VoI baseline's accuracy term.

Single receiver: V2X-Seq-SPD has one CV per sequence. With several receivers, the value becomes a sum over receivers of (8), and the admission problem keeps its form because a broadcast object is sent once.

VIII. Conclusion

CERTISCHED turns the RSU's CPM decision into a receiver-centric, risk-tiered scheduling problem and gives its most critical tier a distribution-free deadline guarantee calibrated online from sidelink measurements, while a virtual queue protects the RSU's own safety traffic. On real perception sequences and real C-V2X service it delivers higher awareness of hidden, conflicting objects than standardized and age-based baselines under the same service contract, at negligible computational cost. Future work will validate the scheduler over the air, extend it to several receivers and to NR-V2X Mode 2.

Data and Code Availability

V2X-Seq-SPD and SEE-V2X are available from their providers [18], [22]. The artifact contains the full pipeline, configuration, tests and all aggregate results; raw data are not redistributed.

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TABLE IV
PRIMARY CONDITION (OUTDOOR INTERSECTION LINKS). SEQUENCE-LEVEL MEANS WITH 95% CLUSTER-BOOTSTRAP INTERVALS; AP OF THE FUSED LIST FOR CAR CLASSES

Policy	Critical awareness	Hidden awareness	Crit. AoI (ms)	AP_{BEV}@0.5	AP_{3D}@0.7	Background	Load (kbit/s)	Cert. miss
[run]								

[run] figure data `fig_alpha_sweep.csv`

Fig. 3. (a) DPP parameter V; (b) certified miss frequency versus target; (c) critical awareness versus CPM size multiplier (primary condition).

TABLE V
CONFIRMATORY TESTS: CERTISCHED MINUS BASELINE (SEQUENCE MEANS, HOLM-ADJUSTED)

Baseline	Metric	Δ	95% CI	Fav.	p_{Holm}
[run]					

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TABLE VI
ABLATIONS (PRIMARY CONDITION)

Variant	Critical awareness	Hidden awareness	Crit. AoI (ms)	$\mathrm{AP}_{\mathrm{BEV}}$@0.5	$\mathrm{AP}_{\mathrm{3D}}$@0.7	Background	Load (kbit/s)	Cert. miss
[run]								

TABLE VII
SECONDARY CONDITION (INDOOR EXTENDED-DURATION LINKS)

Policy	Critical awareness	Hidden awareness	Crit. AoI (ms)	$\mathrm{AP}_{\mathrm{BEV}}$@0.5	$\mathrm{AP}_{\mathrm{3D}}$@0.7	Background	Load (kbit/s)	Cert. miss
[run]								

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